$$\hat{H} = \frac{1}{2m}\left(\hat{p} - \frac{e}{c}\vec{A}\right)^2 - \hat{\mu}\, H + e\varphi$$

vector potential $\vec{A} = (-Hy, 0, 0)$ (Landau gauge)

Recall $\vec{H} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ -Hy & 0 & 0 \end{vmatrix} = \hat{z}\begin{pmatrix} \partial_x & \partial_y \\ -Hy & 0 \end{pmatrix} = H\hat{z}.$

(Any $\vec{A}$ ~~sucks~~ giving such $\vec{H}$ will do)

$$\hat{\mu} = -g\mu_B \hat{S}_z = -\frac{|e|\hbar}{2mc}\hat{\sigma} \quad \text{where} \begin{cases} \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{cases} \text{Pauli matrices}$$

$$\hat{\sigma}|\sigma\rangle = \pm|\sigma\rangle$$

$$\hat{H}\psi_\sigma(\bar{r}) = E\psi_\sigma(\bar{r})$$

$$\left\{\frac{1}{2m}\left(\hat{p}_x + \frac{eH}{c}\hat{y}\right)^2 + \frac{\hat{p}_y^2}{2m} + \frac{\hat{p}_z^2}{2m}\right\}\psi_\sigma(\bar{r}) = \left(E - \mu_B H\sigma\right)\psi_\sigma$$

No dependence on $\hat{x}$ and $\hat{z}$ $\Rightarrow$ $\hat{p}_x$ and $\hat{p}_z$ are conserved

$$\Rightarrow \text{let } \psi_\sigma(\bar{r}) = e^{i(xp_x + zp_z)/\hbar}\,\chi(y)$$

Values: $p_x = \frac{2\pi\hbar}{L_x}N_x, \quad p_z = \frac{2\pi\hbar}{L_z}N_z$

$$\Rightarrow \hat{p}_x e^{i\frac{xp_x}{\hbar}}\chi(y) = -i\hbar\frac{\partial}{\partial x}e^{i\frac{xp_x}{\hbar}}\chi(y) = p_x e^{i\frac{xp_x}{\hbar}}\chi(y)$$

$$\Rightarrow \left\{\frac{1}{2m}\left(p_x + \frac{eH}{c}y\right)^2 + \frac{p_z^2}{2m} + \frac{\hat{p}_y^2}{2m}\right\}\chi(y) = \left(E - \mu_B H\sigma\right)\chi(y)$$

$$\left\{\frac{1}{2m}\left(\frac{eH}{c}\right)^2\left(y+\frac{c}{eH}p_x\right)^2+\frac{\hat{p}_y^2}{2m}\right\}\chi(y)=\left(E-\mu_B H\sigma-\frac{p_z^2}{2m}\right)\chi(y)$$

$$\left\{-\frac{\hbar^2}{2m}\nabla_y^2+\frac{1}{2}m\omega_H^2(y-y_0)^2\right\}\chi=\hat{E}\chi$$

$$y_0=-\frac{cp_x}{eH}$$

Harmonic oscillator, located at y_0: energy spectrum

$$\tilde{E}_n=\hbar\omega_H\left(n+\frac{1}{2}\right),\quad n=0,1,2\ldots$$

$$\downarrow$$

$$E_n-\mu_B H\sigma-\frac{p_z^2}{2m}$$

$\omega_H=\frac{eH}{mc}$ = cyclotron frequency

$$E_n=\underbrace{\hbar\omega_H\left(n+\frac{1}{2}\right)}_{\text{Landau levels}}+\underbrace{\mu_B H\sigma}_{\text{Zeeman energy}}+\underbrace{\frac{p_z^2}{2m}}_{\text{free motion along direction of the field.}}$$

Wavefunction is that of harmonic oscillator.

$$\chi_n(y)=\frac{1}{\pi^{1/4}a_H^{1/2}\sqrt{2^n n!}}\,e^{-\frac{(y-y_0)^2}{2a_H^2}}\,H_n\left(\frac{y-y_0}{a_H}\right)$$

$$a_H=\sqrt{\frac{\hbar}{m\omega_H}}=\sqrt{\frac{\hbar c}{eH}}$$

The center of the orbit $y_0=-\frac{cp_x}{eH}$

is bounded by the sample $0<y_0<L_y$

Although the energy does not depend on y_0!

Number of states $\frac{c}{eH} \frac{2\pi\hbar N_\phi}{L_x} = L_y$

$$N_\phi = \frac{H e L_x L_y}{2\pi\hbar c} = \frac{H S}{\phi_0}$$

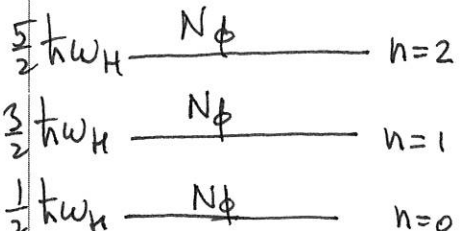
magnetic flux quantum $\phi_0 = \frac{2\pi\hbar c}{e} \approx 4 \cdot 10^{-7}$ Gauss $\cdot$ cm^2

$$N = \frac{\text{Magnetic flux}}{\text{Magnetic flux quantum}}$$

Each ~~magnetic~~ Landau level supports N_ϕ electrons (neglecting spin: if $\mu_B \to 0$ then $2N_\phi$ electrons)

Let us now analyze <u>de Haas-van Alphen oscillations</u> of magnetization within purely 2d model in perpendicular magnetic field: $\varepsilon_n = \hbar\omega_H(n+\frac{1}{2})$, each with degeneracy

$$N_\phi = \frac{eH}{2\pi\hbar c} L_x L_y = \frac{m}{2\pi\hbar^2}\hbar\omega_H (L_x L_y)$$



Assume system with N_e electrons.

Number of states per electron

$$\frac{1}{\nu} = \frac{N_\phi}{N_e} = \frac{H L_x L_y}{N_e \phi_0} = \frac{H}{\rho\phi_0}$$

$$\rho = \frac{N_e}{L_x L_y} = \text{planar density of electrons.}$$

Let's start with very strong magnetic field such that $N_\phi > N_e$ $(0<\nu<1)$

$\Rightarrow$ only 1st Landau level (LL) is occupied:

$$E_0 = N_e \frac{\hbar\omega_H}{2}, \quad \frac{E_0}{N_e} = \frac{1}{2}\hbar\omega_H \Rightarrow \frac{M_0}{N_e} = -\frac{\partial E}{\partial H} = -\frac{\hbar e}{2mc} = -\mu_B$$

$\rightarrow$ now decrease H so that $1<\nu<2$ so that we need to put some electrons ($N_e - N_\phi$) to 2nd LL

$$\Rightarrow E_1 = N_\phi \frac{1}{2}\hbar\omega_H + (N_e - N_\phi)\frac{3}{2}\hbar\omega_H$$

$$\boxed{\hbar\omega_H = 2\mu_B H}$$

$$\frac{E_1}{N_e} = \frac{1}{2}\hbar\omega_H \frac{1}{\nu} + \left(1-\frac{1}{\nu}\right)\frac{3}{2}\hbar\omega_H$$

$$= \left(\frac{3}{2} - \frac{1}{\nu}\right)\hbar\omega_H$$

$$\cancel{\Rightarrow \frac{M_1}{N_e} = -\left(\frac{3}{2}-\frac{1}{\nu}\right)\frac{\hbar e}{mc} = \left(-3+\frac{2}{\nu}\right)\frac{\hbar e}{2mc} = \left(-3+\frac{2}{\nu}\right)\mu_B}$$

But $\frac{1}{\nu} = \frac{H}{\rho \Phi_0}$ depends on H too:

$$\frac{\partial}{\partial H}\left(\frac{E_1}{N_e}\right) = \frac{\partial}{\partial H}\left(3 - \frac{2}{\nu}\right)\mu_B H = \partial_H\left(3\mu_B H - 2\mu_B \frac{H^2}{\rho \Phi_0}\right) =$$

$$= 3\mu_B - 4\mu_B \frac{H}{\rho \Phi_0} = \left(3 - \frac{4}{\nu}\right)\mu_B$$

$$\Rightarrow \frac{M_1}{N_e} = -\frac{\partial E_1}{\partial H} = \left(-3 + \frac{4}{\nu}\right)\mu_B \quad (\text{for } 1<\nu<2)$$

Observe that $\frac{M}{N_e}$ is discontinous:

$$\lim_{\nu \to 1-0} \frac{M}{N_e} = -\mu_B$$

but $$\lim_{\nu = 1+0} \frac{M}{N_e} = (-3+4)\mu_B = +\mu_B$$

Consider the case of $\underline{n < \nu < n+1}$ so that

$$E_n = \sum_{m=0}^{n-1} N_\phi \hbar\omega_H\left(m+\tfrac{1}{2}\right) + \left(N_e - n N_\phi\right)\hbar\omega_H\left(n+\tfrac{1}{2}\right)$$

$$\left.\begin{aligned} &\sum_{m=0}^{n-1} m = 0+1+2+\ldots+n-1 = \tfrac{1}{2}n(n-1) \\ &\sum_{m=0}^{n-1} \tfrac{1}{2} = n\cdot\tfrac{1}{2} \end{aligned}\right\} \quad \tfrac{1}{2}n^2$$

$$\Rightarrow \frac{E_n}{N_e} = \left\{\frac{n^2}{2\nu} + \left(1 - \frac{n}{\nu}\right)\left(n+\tfrac{1}{2}\right)\right\}\hbar\omega_H =$$

$$= \left\{n + \tfrac{1}{2} - \frac{n(n+1)}{2\nu}\right\}\hbar\omega_H$$

$$\frac{n^2}{2\nu} - \frac{n\left(n+\frac{1}{2}\right)}{\nu} = \frac{n^2 - 2\left(n^2 + \frac{1}{2}n\right)}{2\nu} = -\frac{n(n+1)}{2\nu}$$

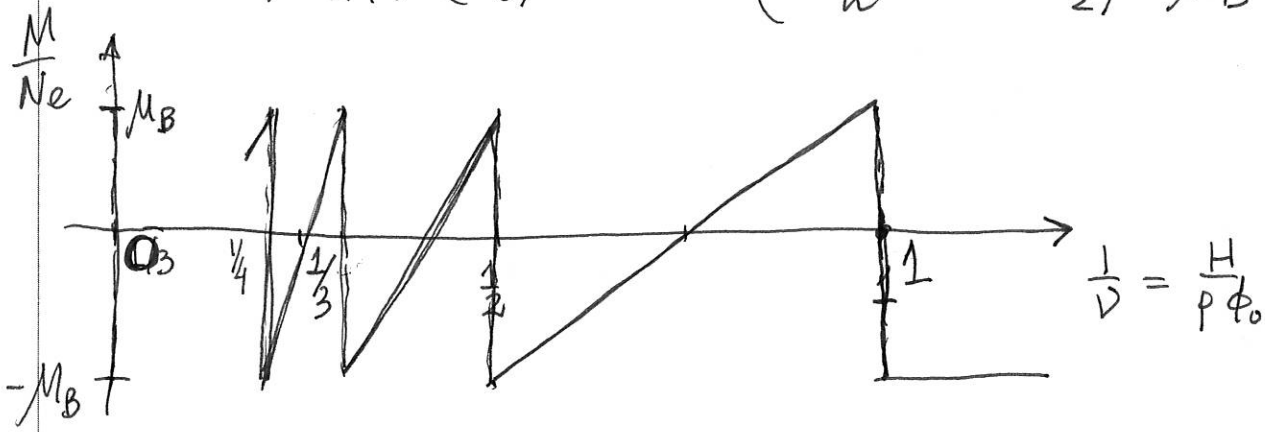
i.e. $\frac{E_n}{N_e} = \left(2n+1 - n(n+1)\frac{H}{p\phi_0}\right)\mu_B H$

$$\frac{\partial M_n/N_e}{\partial H} = -(2n+1)\mu_B + 2n(n+1)\frac{\mu_B H}{p\phi_0} =$$

$$= 2\mu_B\left(\frac{n(n+1)}{\nu} - n - \frac{1}{2}\right) \quad \underline{\text{for } n<\nu<n+1}$$

Thus $\lim_{\nu = n-0}\left(\frac{M}{N_e}\right) = 2\mu_B\left(\frac{(n-1)n}{n} - (n-1) - \frac{1}{2}\right) = -\mu_B$

$$\lim_{\nu = n+0}\left(\frac{M}{N_e}\right) = 2\mu_B\left(\frac{n(n+1)}{n} - n - \frac{1}{2}\right) = \mu_B$$



M goes through discontinuous jumps when $\frac{1}{\nu} = \frac{1}{n}$

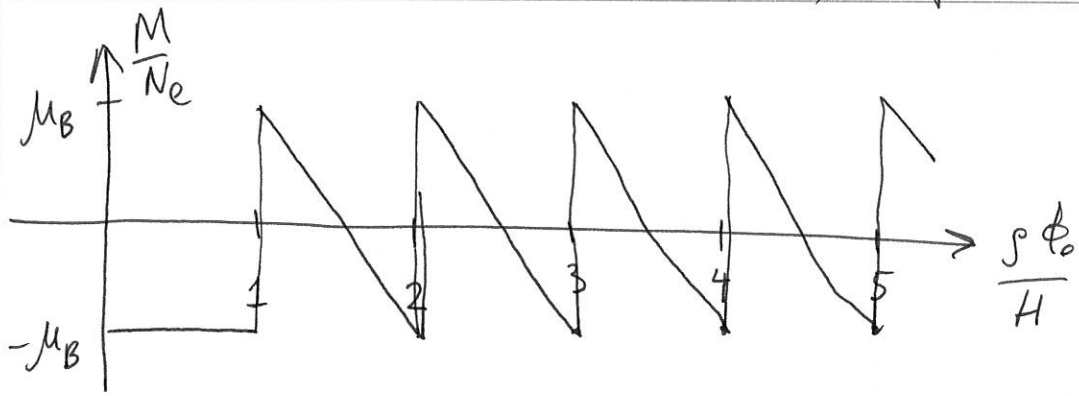
$\Rightarrow$ when $\frac{H}{p\phi_0} = \frac{1}{n} \Rightarrow H_n = \frac{p\phi_0}{n}$

$$\frac{1}{H_n} = \frac{n}{p\phi_0}$$

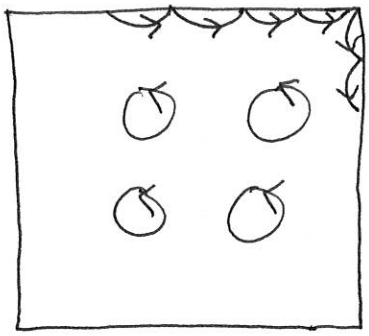
periodic in $\frac{1}{H}$ with period $\frac{1}{p\phi_0}$

$$\frac{1}{H_n} - \frac{1}{H_{n-1}} = \frac{n-(n-1)}{p\phi_0} = \frac{1}{p\phi_0}$$

The same (M/N_e) as function of $\frac{1}{H}$



Landau diamagnetism



Note that skipping orbit results in clockwise rotation, while closed orbits in the bulk are counterclockwise.

Naive classical thinking: $\frac{d\vec{p}}{dt} = \frac{e}{c}\vec{v}\times\vec{B} \Rightarrow m\frac{v^2}{r} = \frac{e}{c}vB$

$$\left(\omega_c = \frac{eB}{mc}\right) \qquad r = \frac{mvc}{eB} = \frac{v}{\omega_c}$$

$\Rightarrow$ orbital moment of the closed orbit

$$\vec{\mu} = \frac{1}{2c}\vec{r}\times\underset{\substack{\uparrow \\ \text{current} \\ \text{density}}}{\vec{J}} \Rightarrow \mu = \frac{evr}{2c} = \frac{1}{2}\frac{mv^2}{B}$$

But this is absurd: 1) independent of charge e.
2) diverges as $B \to 0$

Less naive but still <u>classical</u>:

$$\Omega = -k_BT\ln\left\{\sum_N e^{\mu N/k_BT}\int e^{-E_N(\vec{p},\vec{r})/k_BT}(d\vec{p}\,d\vec{r})\right\}$$

But $E_N = \sum_j \frac{1}{2m}\left(\vec{p}_j - \frac{e}{c}\vec{A}(r_j)\right)^2$ (in magnetic field)

Since integration over $\vec{p}_j$ is unrestricted, can shift $\vec{p} \to \vec{p} + \frac{e}{c}\vec{A}_j$ for any $\vec{A}$ $\Rightarrow$ Ω does not depend on $\vec{A}$ $\Rightarrow$ Ω does not depend on $\vec{B} = \vec{\nabla}\times\vec{A}$

$$\Rightarrow \vec{M} = -\frac{\partial\Omega}{\partial\vec{B}} = 0 \quad \text{(Bohr-van Leeuwen theorem)}$$

Resolution: <u>magnetization</u> is purely <u>quantum</u> effect due to orbit quantization.

Landau (1930)

$$\Omega = -k_B T \underbrace{\left(\frac{BA}{\phi_0}\right)}_{\text{degeneracy}} \sum_n \sum_{p_z,\sigma} \ln\left\{1 + \exp\left(\frac{\mu}{k_B T} - \frac{p_z^2}{2mk_B T} - \frac{\hbar\omega_c}{k_B T}\left(n+\frac{1}{2}\right) - \frac{\mu_B H \sigma}{k_B T}\right)\right\}$$

Note: Ω counts only bulk states here but the bulk ~~distribution~~ contribution is much bigger than surface (due to skipping orbits)

then $M = -\frac{\partial \Omega}{\partial B}$ gives finite result even for an infinite sample.

$$\Omega = -k_B T \, N_\phi \sum_{n, p_z} \ln\left\{1 + \exp\left(\frac{\mu}{k_B T} - \frac{p_z^2}{2mk_BT} - \frac{\hbar\omega_H}{k_BT}(n+\beta_\sigma)\right)\right\}$$

LL degeneracy!

$$\beta_\sigma = \frac{1}{2} + \frac{\sigma \mu_B H}{\hbar\omega_H} = \frac{1}{2} + \sigma \frac{\frac{e\hbar}{2mc} H}{\hbar \frac{eH}{mc}} = \frac{1+\sigma}{2}$$

$$N_\phi = \frac{HS}{\Phi_0}$$

Euler-McLauren f-la: (prove later)

$$\sum_{n=0}^{\infty} f(n+\beta) \approx \int_0^\infty f(y)\,dy + \left(\frac{1}{2}-\beta\right) f(0) + \left(-\frac{1}{12} + \frac{\beta}{2} - \frac{\beta^2}{2}\right) f'(0)$$

Here $f(n+\beta) = \ln\left[1 + e^{\frac{\mu - p_z^2/2m}{k_BT}} \cdot e^{-\frac{\hbar\omega_H}{k_BT}(n+\beta_\sigma)}\right](-k_BT \cdot N_\phi)$

$$\int_0^\infty f(y)\,dy = -k_BT\,N_\phi \int_0^\infty dy \, \ln\left(1 + e^{\frac{\mu - p_z^2/2m - \hbar\omega_H y}{k_BT}}\right)$$

$$= -k_BT\,N_\phi \frac{k_BT}{\hbar\omega_H} \int_0^\infty dx \, \ln\left(1 + e^{\frac{\mu - p_z^2/2m}{k_BT}} \cdot e^{-x}\right)$$

$$x = \frac{\hbar\omega_H}{k_BT} y$$

But $\frac{N_\phi}{\hbar\omega_H} = \frac{HeS}{2\pi\hbar c} \frac{mc}{\hbar eH} = \frac{Sm}{2\pi\hbar^2}$ is H-independent

$\Rightarrow$ this integral term Ω_0 does not contribute to magnetization $M = -\left(\frac{\partial\Omega}{\partial H}\right)_{T,V,\mu}$

Next term, $\left(\frac{1}{2}-\beta\right) f(0) = -\frac{\sigma}{2} f(0)$ will drop Ω_1 out after summing over $\sigma = \pm 1$.

$$N_\phi = \frac{eH}{2\pi\hbar c} L_x L_y = \frac{m}{2\pi\hbar}\left(\frac{eH}{mc}\right) L_x L_y = \frac{m}{2\pi\hbar}\,\omega_H\, L_x L_y$$

Then 3rd term:

$$\Omega_2 = -k_B T\, \frac{m}{2\pi\hbar^2}(\hbar\omega_H)\, L_x L_y \sum_{p_z}\sum_{\sigma}\left(-\frac{1}{12} + \frac{\beta_\sigma}{2} - \frac{1}{2}\beta_\sigma^2\right) f'(0)$$

$$\left(f'(0) = -\frac{\hbar\omega_H}{k_B T}\,\frac{1}{e^{\left(\frac{p_z^2}{2m}-\mu\right)/k_B T}+1} \right)$$

$$= \frac{m}{2\pi\hbar^2}(\hbar\omega_H)^2 L_x L_y \sum_{\sigma, p_z}\left(-\frac{1}{12} + \frac{\beta_\sigma}{2} - \frac{\beta_\sigma^2}{2}\right)\frac{1}{e^{\frac{\varepsilon_z-\mu}{k_B T}}+1}$$

$$\left(\varepsilon_z \equiv \frac{p_z^2}{2m}\right)$$

$$\sum_\sigma\left(-\frac{1}{12} + \frac{\beta_\sigma}{2}\right) = -\frac{1}{6} + \sum_\sigma\left(\frac{1}{4} + \frac{\sigma\mu_B H}{2\hbar\omega_H}\right) = -\frac{1}{6} + \frac{1}{2}.$$

$$-\frac{1}{2}\sum_\sigma \beta_\sigma^2 = -\frac{1}{2}\sum_\sigma\left(\frac{1}{4} + \left(\frac{\sigma\mu_B H}{\hbar\omega_H}\right)^2 + \frac{\sigma\mu_B H}{\hbar\omega_H}\right)$$

$$= -\frac{1}{2}\left(\frac{1}{2} + 2\left(\frac{\mu_B H}{\hbar\omega_H}\right)^2\right) = -\frac{1}{4} - \left(\frac{\mu_B H}{\hbar\omega_H}\right)^2$$

Thus $$\Omega_2 = -\frac{m L_x L_y}{2\pi\hbar^2}(\hbar\omega_H)^2 \sum_{p_z}\left[\left(\frac{\mu_B H}{\hbar\omega_H}\right)^2 - \frac{1}{12}\right] n_F(\varepsilon_z)$$

$$+\frac{1}{6} - \frac{1}{2} + \frac{1}{4} = \frac{1}{6} - \frac{1}{4} = \frac{2-3}{12} = -\frac{1}{12}$$

~~Third term: $\sum_\sigma \left(-\frac{1}{12} + \frac{\beta}{2} - \frac{\beta^2}{2}\right) f'(0) = \sum_\sigma \left(-\frac{1}{12} + \frac{1+\sigma}{4} - \frac{(1+\sigma)^2}{8}\right) f'(0)$~~

~~$= \left(-\frac{1}{6} + \frac{1}{2} - \frac{1}{2}\right) f'(0) = -\frac{1}{6} f'(0)$~~

$$f'(0) = \frac{\partial}{\partial x} \ln\left(1 + e^{\frac{\mu - p_z^2/2m}{k_B T}} e^{-x}\right)\Bigg|_{x=0} \cdot \underbrace{\frac{\partial x}{\partial y}}_{\frac{\hbar\omega_H}{k_B T}} =$$

$$= \frac{\hbar\omega_H}{k_B T} \frac{-e^{\frac{\mu - p_z^2/2m}{k_B T}} e^{-x}}{1 + e^{\frac{\mu - p_z^2/2m}{k_B T}} e^{-x}}\Bigg|_{x=0} =$$

$$= -\frac{\hbar\omega_H}{k_B T} \frac{1}{e^{x} e^{\frac{p_z^2/2m - \mu}{k_B T}} + 1}\Bigg|_{x=0}$$

$$= -\frac{\hbar\omega_H}{k_B T} \frac{1}{e^{\left(\frac{p_z^2}{2m} - \mu\right)/k_B T} + 1}$$

~~$\Delta S = \Delta S_0 - k_B T N_\phi \left(\frac{1}{6} \frac{\hbar\omega_H}{k_B T}\right) \frac{\partial}{\partial p_z} \frac{1}{e^{\left(\frac{p_z^2}{2m} - \mu\right)/k_B T} + 1}$~~

$$\Omega_2 = -\frac{m L_x L_y L_z}{2\pi\hbar^2}\int \frac{dp_z}{2\pi\hbar}\left(\mu_B^2 H^2 - \frac{(\hbar\omega_H)^2}{12}\right) n_F\left(\frac{p_z^2}{2m}\right)$$

where $n_F(x) \equiv \dfrac{1}{e^{\frac{x-\mu}{k_B T}}+1}$ is Fermi distribution function.

Magnetic moment $M = -\dfrac{\partial \Omega}{\partial H}$

Magnetic susceptibility

$$\chi = \frac{\partial M}{\partial H} = 2\mu_B^2 \frac{\overset{\text{volume}}{V} m^*}{2\pi\hbar^2}\left(1 - \frac{1}{12}\left(\frac{\hbar\omega_H}{H\mu_B}\right)^2\right) \cdot$$

$$\cdot \int \frac{dp_z}{2\pi\hbar}\, n_F\left(\frac{p_z^2}{2m}\right)$$

$$\frac{\hbar\omega_H}{H\mu_B} = \frac{\hbar e H}{\underset{\text{orbital mass}}{m_e}} \cdot \frac{2\overset{\text{bare mass}}{m_e}}{e\hbar H} = 2\frac{m_e}{m^*}$$

Per unit volume

$$\frac{\chi}{V} = 2\mu_B^2 \frac{m^*}{2\pi\hbar^2}\left[1 - \frac{1}{3}\left(\frac{m_e}{m^*}\right)^2\right] \cdot \int \frac{dp_z}{2\pi\hbar}\, n_F\left(\frac{p_z^2}{2m}\right)$$

Pauli paramagnetism (from Zeeman interaction with mag. field)

Landau diamagnetism (from orbital motion of electrons)

At $T=0$: $$\int \frac{dp_z}{2\pi\hbar}\, n_F\left(\frac{p_z}{2m}\right) = \int_{-p_F}^{p_F} \frac{dp_z}{2\pi\hbar} = \frac{2p_F}{2\pi\hbar}$$

$$\frac{\chi}{V} = \mu_B^2 \frac{m^* p_F}{\pi^2\hbar^3}\left[1 - \frac{1}{3}\left(\frac{m_e}{m^*}\right)^2\right] = \mu_B^2 \nu_F \left[1 - \frac{1}{3}\left(\frac{m_e}{m^*}\right)^2\right]$$

But $$\frac{m^* p_F}{\pi^2\hbar^3} = \frac{\overset{2\text{ (spin degeneracy)}}{g}\, m^{3/2}}{\sqrt{2}\,\pi^2\hbar^3}\sqrt{\varepsilon_F} = \text{DoS at Fermi level} = \nu_F$$

$$\varepsilon_F = \frac{p_F^2}{2m}$$

Start with identity $\sum_{n=-\infty}^{+\infty} \delta(x-n) = \sum_{k=-\infty}^{\infty} e^{i2\pi kx}$

Multiply both sides by $\int_{-\infty}^{\infty} dx\, F(x) \Rightarrow$

$$\Rightarrow \sum_n F(n) = \sum_{k=-\infty}^{+\infty} \int_{-\infty}^{\infty} dx\, F(x)\, e^{i2\pi kx}$$

but $\int_0^{\infty} dx\, F(x)\, e^{i2\pi kx} = \int d\left(\frac{e^{i2\pi kx}}{i2\pi k}\right) F(x) =$

$$= \frac{e^{i2\pi kx} F(x)}{i2\pi k}\Big|_0^{\infty} - \int_0^{\infty} F'(x) \frac{e^{i2\pi kx}}{i2\pi k} dx$$

$$= -\frac{F(0)}{i2\pi k} - \int_0^{\infty} dx\, F'(x)\, e^{i2\pi kx}$$

Assuming $F(x=\infty) = 0 = F'(x=\infty) = F^{(n)}(x=\infty)$.

$$\Rightarrow \sum_{n=-\infty}^{+\infty} F(n) = \underbrace{\int_{-\infty}^{\infty} dx\, F(x)}_{k=0 \text{ term}} + \sum_{k=-\infty}^{-1} \int_{-\infty}^{\infty} dx\, F(x)\, e^{+i2\pi kx}$$

$$+ \sum_{k=1}^{\infty} \int_{-\infty}^{\infty} dx\, F(x)\, e^{i2\pi kx}$$

Assume <u>even</u> and real function $F(x) = F(-x)$ (note that for <u>odd</u> function $\sum_{n=-\infty}^{+\infty} F(n) = 0$ anyway)

$$\Rightarrow F(0) + 2\sum_{n=1}^{\infty} F(n) = 2\int_0^{\infty} dx\, F(x) +$$

$$+ \int_{-\infty}^{\infty} dx \sum_{k=1}^{\infty} F(x) \cdot 2\cos(2\pi kx) = 2\int_0^{\infty} dx\, F(x) +$$

$$+ 4\sum_{k=1}^{\infty} \int_0^{\infty} dx\, F(x) \cos(2\pi kx)$$

$\Rightarrow$ divide both sides by 2:

$$\frac{1}{2}F(0) + \sum_{n=1}^{\infty} F(n) = \int_0^{\infty} dx\, F(x) + 2\,\mathrm{Re}\left\{ \sum_{k=1}^{\infty} \int_0^{\infty} dx\, F(x) \cdot e^{i2\pi kx} \right\}$$

Now, integrating by parts (and assuming that $F^{(n)}(x=\infty)=0$ for all $n=0,1,2\ldots$) we find

$$\int_0^{\infty} dx\, F(x)\, e^{i2\pi kx} = -\frac{F(0)}{2\pi k i} + \frac{F'(0)}{(2\pi k i)^2} - \frac{F^{(2)}(0)}{(2\pi k i)^3} + \frac{F^{(3)}(0)}{(2\pi k i)^4} + \ldots$$

Taking <u>real part</u> and using $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$, $\sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{\pi^4}{90}$

we find

$$\frac{1}{2}F(0) + \sum_{n=1}^{\infty} F(n) = \int_0^{\infty} dx\, F(x) - \frac{1}{12}F'(0) + \frac{1}{720}F^{(3)}(0) + \ldots$$

OR, $\sum_{n=0}^{\infty} F(n) = \int_0^{\infty} dx\, F(x) + \frac{1}{2}F(0) - \frac{1}{12}F'(0) + \frac{1}{720}F^{(3)}(0) + \ldots$

<u>Next</u>: little thinking shows that coefficients of the expansion ($-\frac{1}{12}$, $\frac{1}{720}$, etc) are determined by the oscillating term <u>$e^{i2\pi kx}$</u> only, and are universal for <u>all</u> functions.

This allows us to solve for more general relation:

$$\sum_{n=0}^{\infty} f(\beta+n) \approx \int_0^{\infty} dx\, f(x) + a f(0) + b f'(0) + c f''(0) + \dots$$

where offset β is $|\beta| < 1$.

To find $a, b, c, \dots$ we then can choose some <u>convenient</u> function and calculate them for it!

Let's choose $f(\beta+n) = e^{-(n+\beta)y}$ (y = parameter here)

But $\sum_{n=0}^{\infty} e^{-(n+\beta)y} = e^{-\beta y} \sum_{n=0}^{\infty} e^{-ny} = \frac{e^{-\beta y}}{1-e^{-y}} =$

$$= (\text{expand in powers of } y) = \frac{1-\beta y + \frac{1}{2!}(\beta y)^2 + \dots}{y - \frac{1}{2!}y^2 + \frac{1}{3!}y^3 + \dots} \approx$$

$$\approx \frac{1}{y} + \left(\frac{1}{2} - \beta\right) + \left(\frac{1}{12} - \frac{\beta}{2} + \frac{\beta^2}{2}\right) y + \left(-\frac{\beta}{12} + \frac{\beta^2}{4} - \frac{\beta^3}{6}\right) y^2 + \dots$$

On the other hand, $\int_0^{\infty} dx\, e^{-xy} = \frac{1}{y}$

$$\Rightarrow \frac{1}{y} + \left(\frac{1}{2} - \beta\right) + \left(\frac{1}{12} - \frac{\beta}{2} + \frac{\beta^2}{2}\right) y + \left(-\frac{\beta}{12} + \frac{\beta^2}{4} - \frac{\beta^3}{6}\right) y^2 \dots$$

$$= \frac{1}{y} + a + by + cy^2 + \dots \Rightarrow \begin{cases} a = \frac{1}{2} - \beta \\ b = -\frac{1}{12} + \frac{\beta}{2} - \frac{\beta^2}{12} \\ c = -\frac{\beta}{12} + \frac{\beta^2}{4} - \frac{\beta^3}{6} \end{cases}$$

$$f(0) = e^{-(n+\beta)y}\Big|_{n+\beta=0} = 1$$

$$f'(0) = \frac{d}{dn} e^{-(n+\beta)y}\Big|_{n+\beta=0} = -y\, e^{-(n+\beta)y}\Big|_{n+\beta=0} = -y$$

$$f''(0) = \frac{d^2}{dn^2} e^{-(n+\beta)y}\Big|_{n+\beta=0} = y^2.$$

$\Rightarrow$ hence $\displaystyle\sum_{n=0}^{\infty} f(n+\beta) \approx \int_0^{\infty} dx\, f(x) + \left(\frac{1}{2}-\beta\right) f(0)$

will use it later

$$+\left(-\frac{1}{12}+\frac{\beta}{2}-\frac{\beta^2}{12}\right) f'(0) + \left(-\frac{\beta}{12}+\frac{\beta^2}{4}-\frac{\beta^3}{6}\right) f''(0) + \ldots$$

Observe that for special case $\beta=0$ we get precisely the set of previous coefficients:

$$\sum_{n=0}^{\infty} f(n) = \int_0^{\infty} dx\, f(x) + \frac{1}{2} f(0) - \frac{1}{12} f'(0) + (0) f''(0) + \ldots$$